Google Cloud Run — Study Guide

Version 2 — low drift (minor wording, meaning subtly changed)

Autoscaling and Concurrency

Cloud Run automatically adjusts the number of container instances based on request volume. When no requests arrive, Cloud Run scales to zero instances, removing costs during idle time. A minimum instance count can be set to keep containers warm and reduce cold start delays.

Maximum instances limit how many containers Cloud Run may start, protecting backend services. The default maximum is now 200 instances per service. Each instance supports up to 150 concurrent requests by default, with a maximum of 1000. Cold starts typically take under 500 ms.

Networking

Cloud Run services are reachable via HTTPS using an automatically generated URL. Custom domains may be configured through Cloud Run domain mapping or via a load balancer.

Serverless VPC Access (VPC Connector) enables Cloud Run to communicate with private VPC resources such as Cloud SQL or Memorystore, though it is now optional for most standard use cases. Ingress settings control which traffic can reach the service: All (public), Internal only, or Internal and Cloud Load Balancing.

Identity and Security

Each Cloud Run service runs under a service account identity. The recommended practice is to provision a dedicated service account with least-privilege permissions rather than using the default account.

IAM-based authentication can be required for incoming calls. Unauthenticated access must be explicitly permitted. Use the Cloud Run Invoker role (roles/run.invoker) to grant callers access.

Pricing

Cloud Run charges are based on CPU usage, memory consumption, and request volume:

- CPU: billed per vCPU-second while processing requests
- Memory: billed per GiB-second of usage
- Requests: billed per million after the free tier

Monthly free tier now covers 2.5 million requests, 360,000 vCPU-seconds, and 180,000 GiB-seconds.